Boost converter (12 V)

L1 MWSA0402S-100MT

U1 TLV61046A (SW, VIN, GND, VO, FB, EN)

C2 4.7u, C3 10u

U1 TLV61046A boost from +5V (USB VBUS). FB and EN tied to VIN: internal fixed 12 V output, always enabled.
L1 10 uH, C2 4.7u in, C3 10u out.
12 V out feeds the Q2 keying switch.

Keying

TX (MCU PA4) drives Q1 2N7002; R1 22k pull-down holds it off at reset.
Q1 pulls Q2 MMBT4403 base via R3 22k; high-side switch for the 12 V PA supply. R2 22k B-E pull-up, C4 10n C-B.
R6 10R + C7 1u filter.
R8 1R 1% PA current sense: measure across J2 PAI+ / J3 PAI-.

Q2 MMBT4403, R2 22k, R3 22k, C4 10n, R6 10, R8 1 1%, C7 1u, J2 PAI+, J3 PAI-, Q1 2N7002, R1 22k, TX

Power amplifier

CLK0 (SI5351C) -> U3 SN74LVC1G17 buffer -> C5 100n -> Q3/Q8 BS170 in parallel (DNP, hand-solder), R16/R17 10R gate stoppers.
Drain fed from J3 via L2. R5 3.3k + C8 100n: drain to buffer-out RC. C11 100n DC block to LPF_IN.

U3 SN74LVC1G17, C1 100n, C5 100n, R5 3.3k, C8 100n, C11 100n, L2 FB2402-61 5T, R16 10, R17 10, R7 56, Q3 BS170, Q8 BS170, CLK0

PA bias

U5 MCP4725 12-bit DAC on SCL_5V / SDA_5V (A0 = GND).
VOUT -> R7 56R -> PA gates.
C9 1u on VOUT, C10 100n VDD.
J1: bias voltage test point.

MCP4725A1T-E/CH U5 (VOUT A0, VSS SCL, VDD SDA), J1 Bias V, C9 1u, C10 100n, SCL_5V, SDA_5V

Si5351C synthesizer

R4 10, C6 100n, U2 25.000 MHz TCXO, C12 100n, C14 100n, C50 100n, C51 100n, C52 100n, C53 100n

J16 SMA CLKIN, CLKIN_SMA, C54 100n, D21 PESD5V0F1BL, R43 49.9, R44 10k, R45 10k, R46 10k, SI_INTR

U4 SI5351C-B-GMR (XA, XB, CLKIN, SDA, SCL, INTR, OEB, VDD, VDDOA, VDDOB, VDDOC, VDDOD, GND, CLK0, CLK1, CLK2, CLK3, CLK4, CLK5, CLK6, CLK7)

J6 SMA, R47 33, CAL_CLK, CLK0

XA: U2 25 MHz TCXO, R4 10R feed, R13 OR OE (DNP).
C12 100n to XA; XB open.
J16 CLKIN (10-100 MHz): R43 49.9R termination, D21 PESD5V0F1BL ESD, C54 100n AC coupling.
R44/R45 10k/10k bias at VDD/2.
OEB tied to GND; outputs enabled over I2C.

RF output: low-pass filter bank (sheet 3)

Filter Bank (BAND_A, BAND_B, LPF_IN, LPF_OUT)

File: filter_bank.kicad_sch

BAND_A/BAND_B select one of four LPFs:
00 80/60 m, 01 40/30 m, 10 20/17 m, 11 15/12/10 m.
J11 BNC output (DNP, hand-soldered).

J11 BNC

MCU + USB

File: mcu_usb.kicad_sch

GPS

File: gps.kicad_sch

I/O + UI

File: io_ui.kicad_sch

J9 GND

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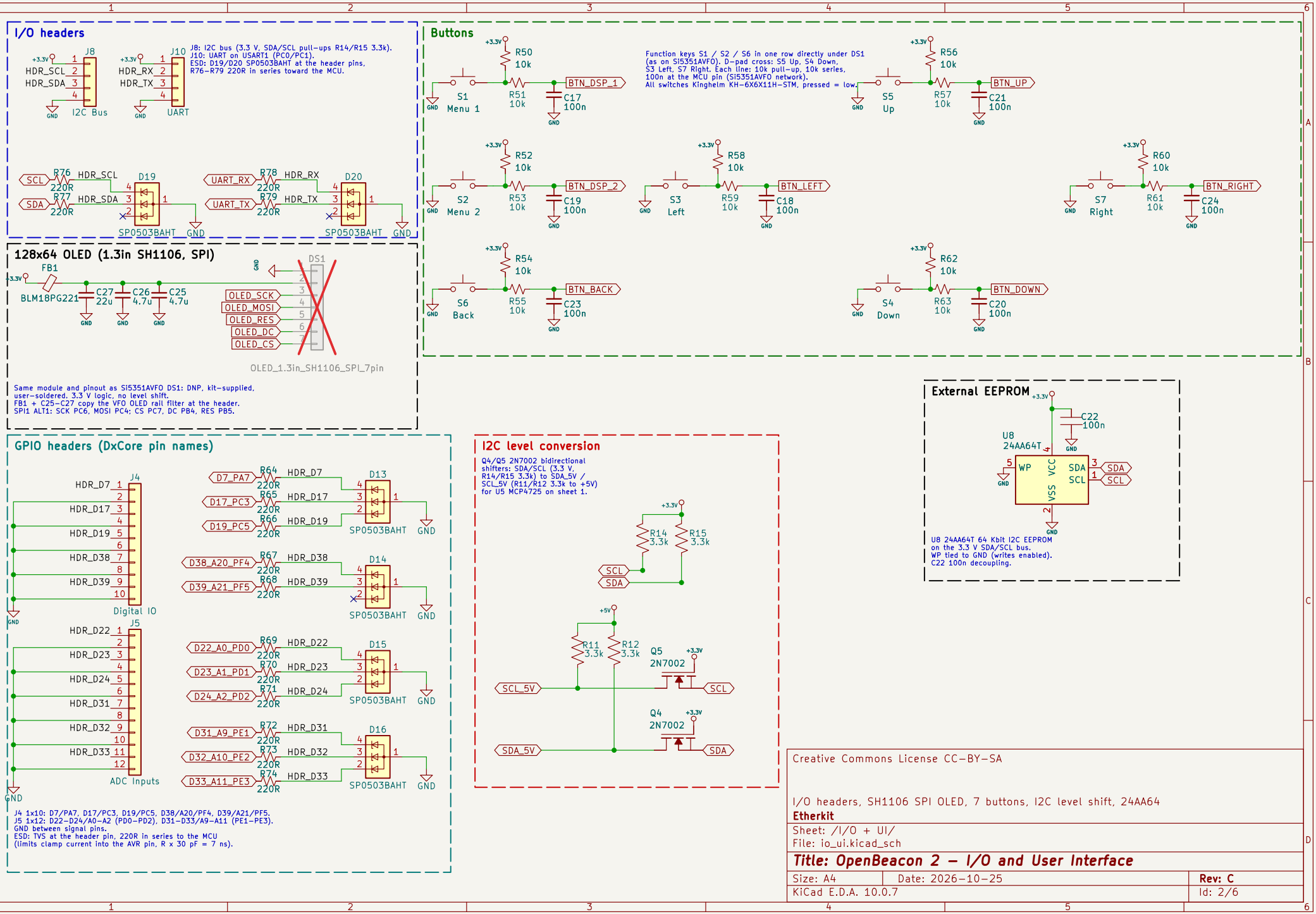
RF chain: boost, keying, PA, bias DAC, Si5351C synth, filter bank, BNC out

Etherkit

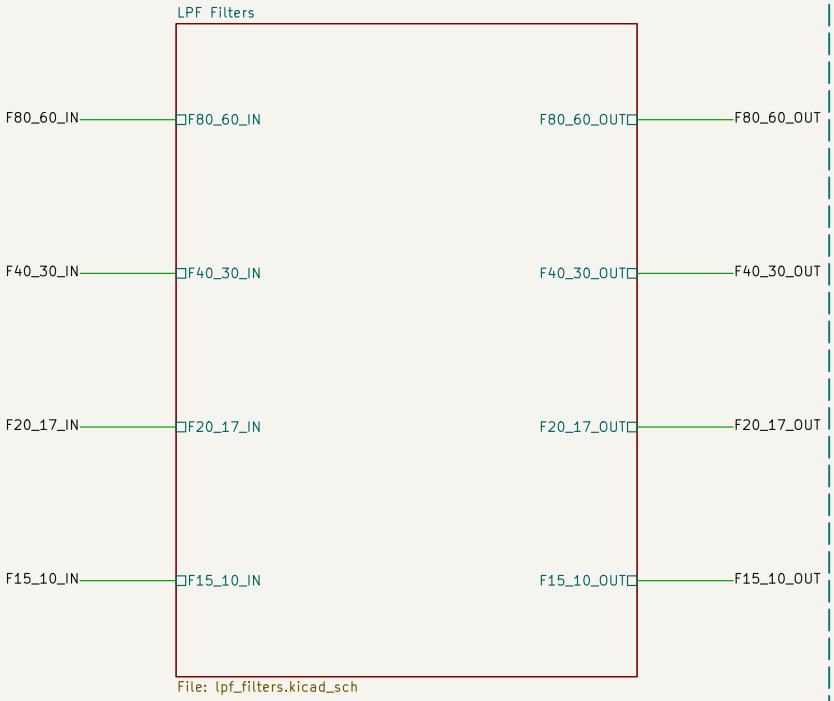
Sheet: /
File: OpenBeacon2.kicad_sch

Title: OpenBeacon 2

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 1/6

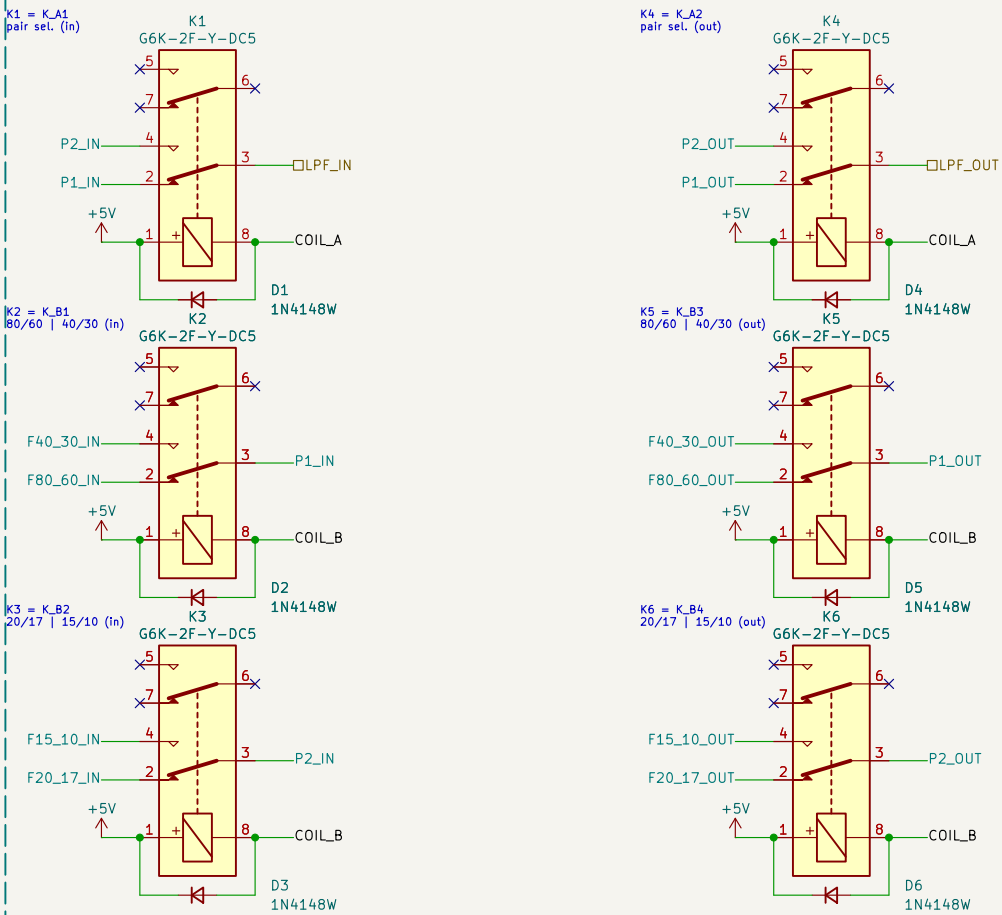

I/O headers
J8: I2C bus (3.3 V, SDA/SCL pull-ups R14/R15 3.3k).
J10: UART on USART1 (PC0/PC1).
ESD: D19/D20 SP0503BAHT at the header pins,
R76–R79 220R in series toward the MCU.
J8
J10
HDR_SCL
HDR_SDA
HDR_RX
HDR_TX
I2C Bus
UART
SCL
SDA
UART_RX
UART_TX
R76 220R
R77 220R
R78 220R
R79 220R
D19
D20
SP0503BAHT
128x64 OLED (1.3in SH1106, SPI)
FB1
BLM18PG221
C27 22u
C26 4.7u
C25 4.7u
DS1
OLED_SCK
OLED_MOSI
OLED_RES
OLED_DC
OLED_CS
OLED_1.3in_SH1106_SPI_7pin
Same module and pinout as SI5351AVFO DS1: DNP, kit-supplied,
user-soldered, 3.3 V logic, no level shift.
FB1 + C25–C27 copy the VFO OLED rail filter at the header.
SPI1 ALT1: SCK PC6, MOSI PC4; CS PC7, DC PB4, RES PB5.
Buttons
Function keys S1 / S2 / S6 in one row directly under DS1
(as on SI5351AVFO). D-pad cross: S5 Up, S4 Down,
S3 Left, S7 Right. Each line: 10k pull-up, 10k series,
100n at the MCU pin (SI5351AVFO network).
All switches Kinghelm KH-6X6X11H-STM, pressed = low.
S1 Menu 1
S2 Menu 2
S3 Left
S4 Down
S5 Up
S6 Back
S7 Right
R50 10k
R51 10k
R52 10k
R53 10k
R54 10k
R55 10k
R56 10k
R57 10k
R58 10k
R59 10k
R60 10k
R61 10k
R62 10k
R63 10k
C17 100n
C18 100n
C19 100n
C20 100n
C21 100n
C22 100n
C23 100n
C24 100n
BTN_DSP_1
BTN_DSP_2
BTN_BACK
BTN_LEFT
BTN_UP
BTN_DOWN
BTN_RIGHT
External EEPROM
U8
24AA64T
WP
VCC
VSS
SDA
SCL
U8 24AA64T 64 Kbit I2C EEPROM
on the 3.3 V SDA/SCL bus.
WP tied to GND (writes enabled).
C22 100n decoupling.
GPIO headers (DxCore pin names)
J4
J5
Digital IO
ADC Inputs
HDR_D7
HDR_D17
HDR_D19
HDR_D38
HDR_D39
HDR_D22
HDR_D23
HDR_D24
HDR_D31
HDR_D32
HDR_D33
D7_PA7
D17_PC3
D19_PC5
D38_A20_PF4
D39_A21_PF5
D22_A0_PD0
D23_A1_PD1
D24_A2_PD2
D31_A9_PE1
D32_A10_PE2
D33_A11_PE3
R64 220R
R65 220R
R66 220R
R67 220R
R68 220R
R69 220R
R70 220R
R71 220R
R72 220R
R73 220R
R74 220R
D13
D14
D15
D16
J4 1x10: D7/PA7, D17/PC3, D19/PC5, D38/A20/PF4, D39/A21/PF5.
J5 1x12: D22–D24/A0–A2 (PD0–PD2), D31–D33/A9–A11 (PE1–PE3).
GND between signal pins.
ESD: TVS at the header pin, 220R in series to the MCU
(limits clamp current into the AVR pin, R x 30 pF = 7 ns).
I2C level conversion
Q4/Q5 2N7002 bidirectional
shifters: SDA/SCL (3.3 V
R14/R15 3.3k) to SDA_5V /
SCL_5V (R11/R12 3.3k to +5V)
for U5 MCP4725 on sheet 1.
R14 3.3k
R15 3.3k
R11 3.3k
R12 3.3k
Q5 2N7002
Q4 2N7002
SCL_5V
SDA_5V
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I/O headers, SH1106 SPI OLED, 7 buttons, I2C level shift, 24AA64
Etherkit
Sheet: /I/O + UI/
File: io_ui.kicad_sch
Title: OpenBeacon 2 – I/O and User Interface
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 2/6

F2–F5 low-pass filters (T30 toroids)

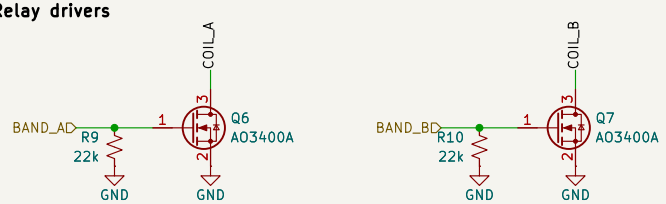


Four 7th-order Cauer LPFs, hand-wound T30 toroids.
Schematic + winding table on sheet LPF Filters.
Designed for 1 W output max.

Relay tree (2 levels, input + output mirrored; all off = 80/60 m)



Relay drivers



BAND_A -> K1/K4, BAND_B -> K2/K3/K5/K6.
Q6 (BAND_A): 2 coils, 42 mA nom (5 V / 237 R),
<= 49 mA at 5.25 V, -10 % coil R.
Q7 (BAND_B): 4 coils, 84 mA nom, <= 98 mA.
AO3400A: 5.7 A cont., 48 mR @ Vgs 2.5 V -> <5 mV drop.
One 1N4148W flyback per coil, at each relay.

Band code + notes

Band code (BAND_A, BAND_B):
0 0 80/60 m (all relays off, NC path, default)
0 1 40/30 m (K2, K3, K5, K6 on)
1 0 20/17 m (K1, K4 on)
1 1 15/12/10 m (all six on)
Unselected filters float (open at both ends).
+5V relay load: 0 / 84 / 42 / 127 mA nominal (<=147 mA).
Relay pins: COM right, NC lower left, NO upper left; pole 2 unused.
Input and output paths never share a relay body.
Layout: keep K1–K2–K3 and K4–K5–K6 traces short,
50 ohm CPWG over solid GND, filters next to their relays.

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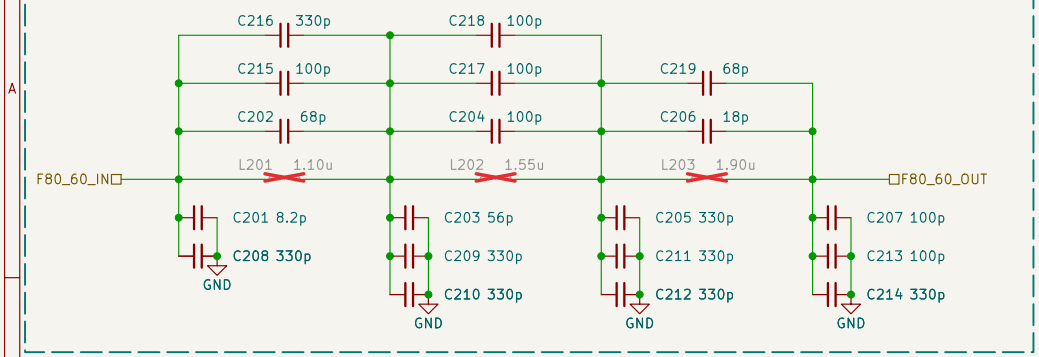
4 7th-order Cauer LPFs, 6-relay 2-level tree (BAND_A/BAND_B)

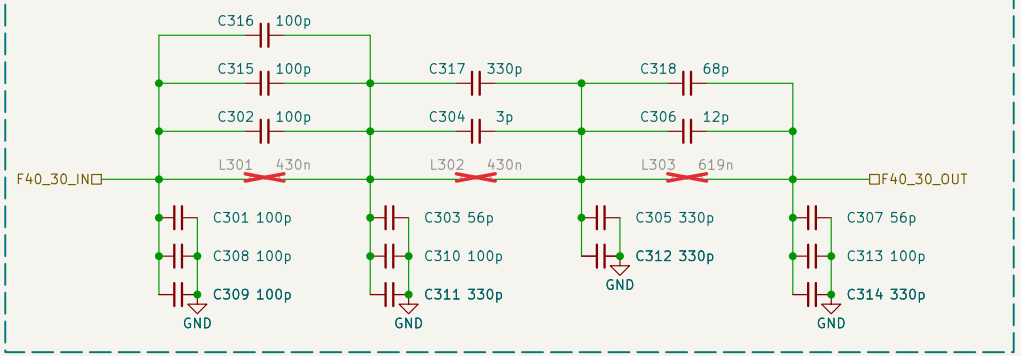
Etherkit

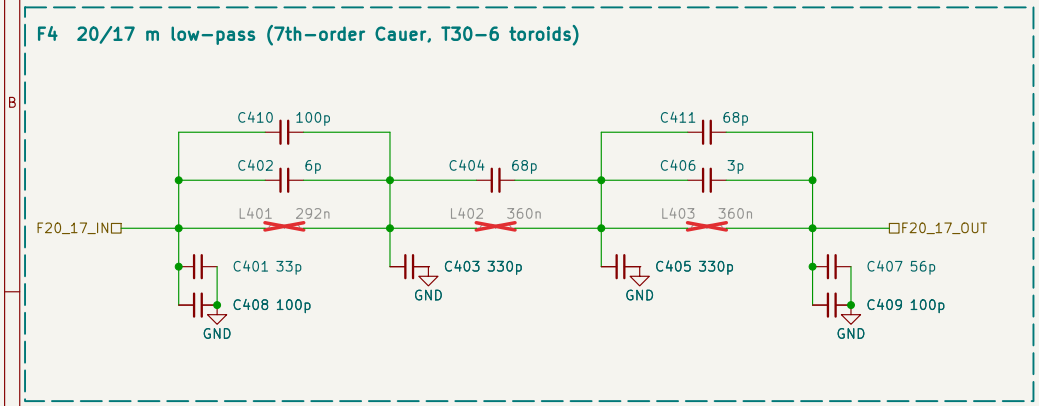
Sheet: /Filter Bank/
File: filter_bank.kicad_sch

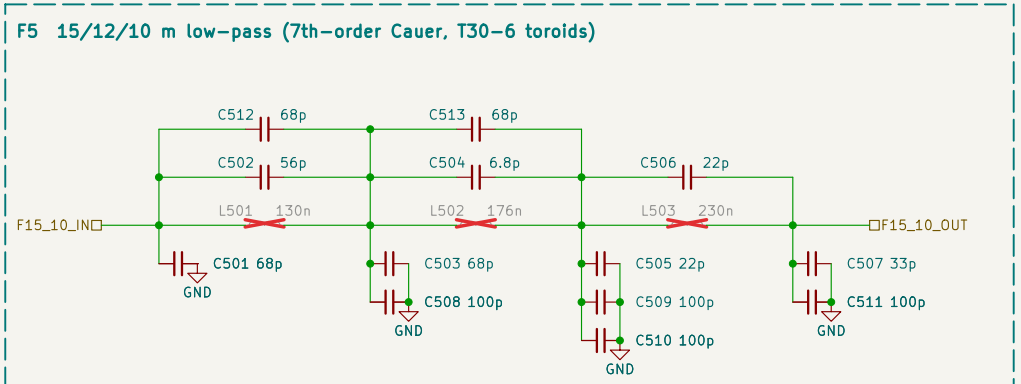
Title: OpenBeacon 2 – Low-Pass Filter Bank

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 3/6









Toroid winding table (hand-wound, not placed by JLC)

Ref	Filter / bands	Inductance	Core (Kits and Parts)	Turns	Wire
L201	F2 80/60 m	1.10 uH	T30-2 (red)	16	#28 AWG enam.
L202	F2 80/60 m	1.55 uH	T30-2 (red)	19	#28 AWG enam.
L203	F2 80/60 m	1.90 uH	T30-2 (red)	21	#30 AWG enam.
L301	F3 40/30 m	430 nH	T30-2 (red)	10	#24 AWG enam.
L302	F3 40/30 m	430 nH	T30-2 (red)	10	#24 AWG enam.
L303	F3 40/30 m	619 nH	T30-2 (red)	12	#26 AWG enam.
L401	F4 20/17 m	292 nH	T30-6 (yellow)	9	#24 AWG enam.
L402	F4 20/17 m	360 nH	T30-6 (yellow)	10	#24 AWG enam.
L403	F4 20/17 m	360 nH	T30-6 (yellow)	10	#24 AWG enam.
L501	F5 15/12/10 m	130 nH	T30-6 (yellow)	6	#24 AWG enam.
L502	F5 15/12/10 m	176 nH	T30-6 (yellow)	7	#24 AWG enam.
L503	F5 15/12/10 m	230 nH	T30-6 (yellow)	8	#24 AWG enam.

Cores: Micrometals T30-2 (red, F2/F3) and T30-6 (yellow, F4/F5) from Kits and Parts (kitsandparts.com).
Turns = passes through the core centre. AL 4.3 nH/t^2 (-2), 3.6 nH/t^2 (-6), +/-5%%.
F5 has few turns: tune each toroid by spreading or compressing turns. Mount toroids vertically.
Rated for 1 W output max. All filter caps 0603 C0G/NP0 50 V; parallel parts sum to the design value.

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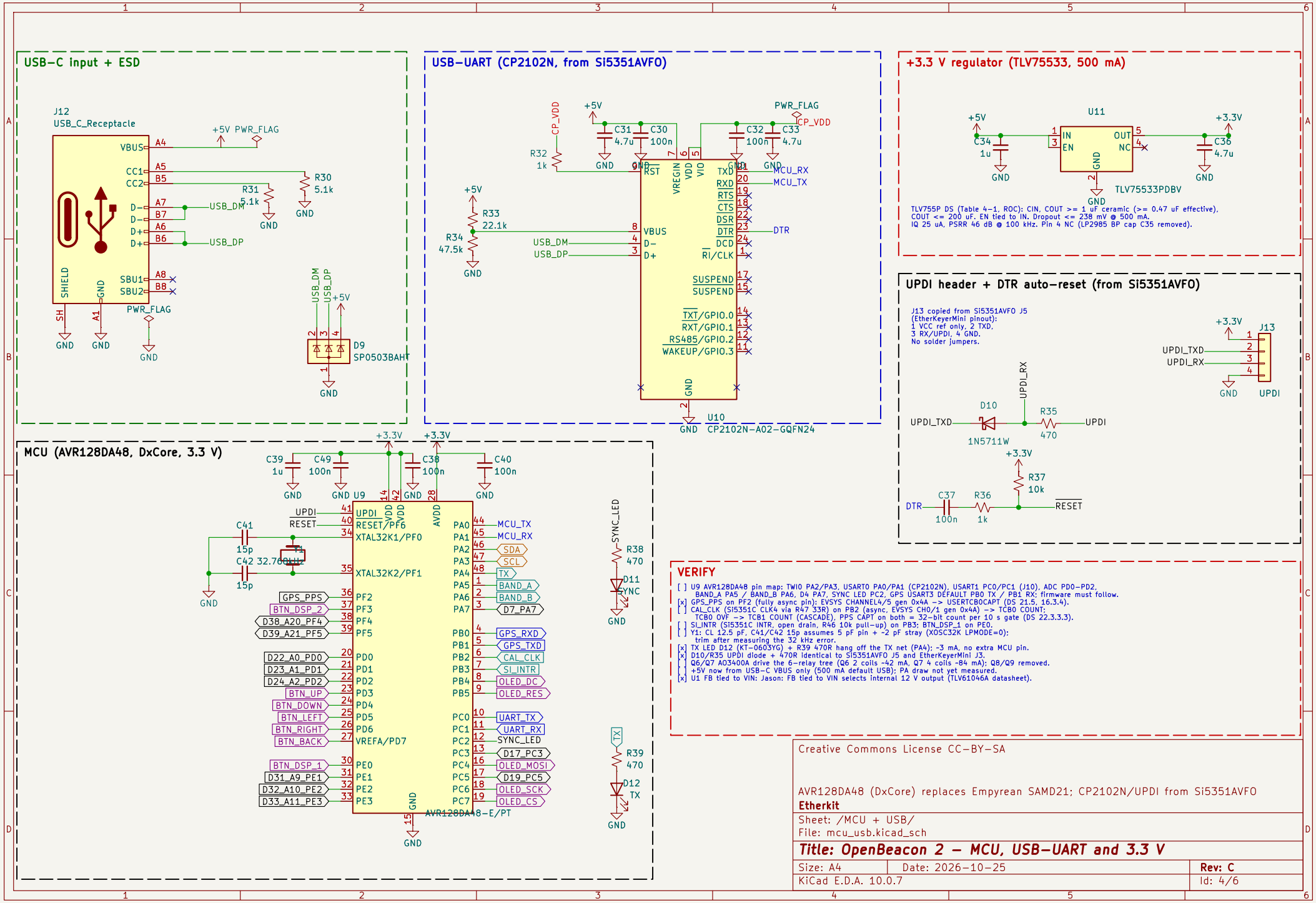
F2-F5 7th-order Cauer LPFs on hand-wound T30 toroids, 1 W max

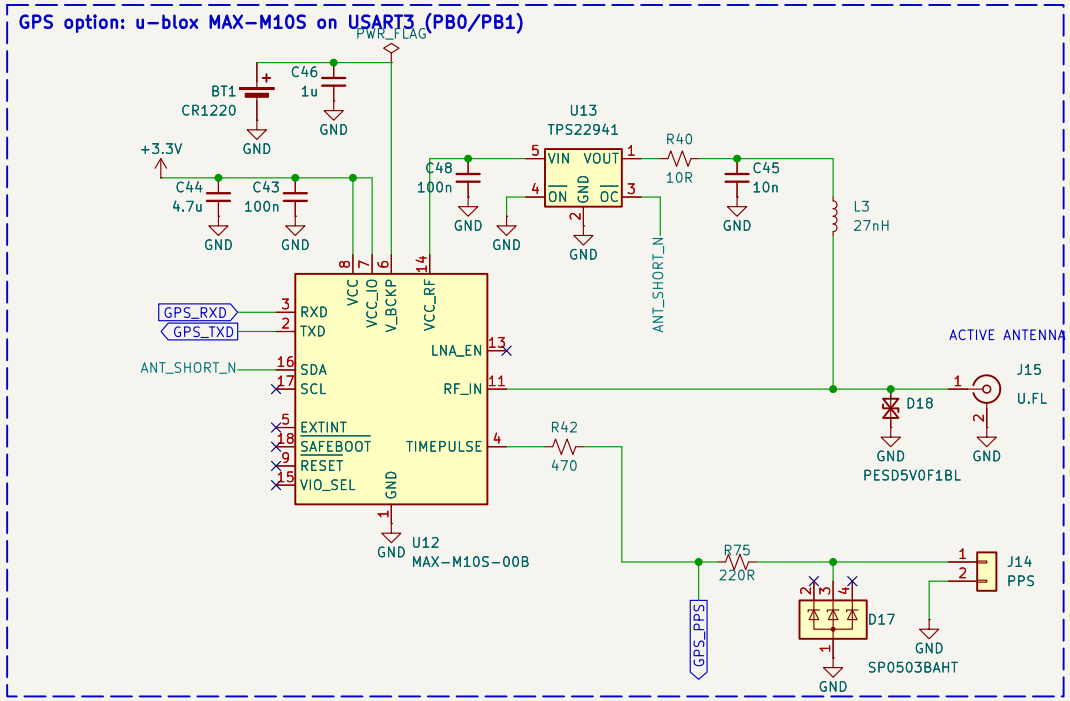
Etherkit

Sheet: /Filter Bank/LPF Filters/
File: lpf_filters.kicad_sch

Title: OpenBeacon 2 - LPFs F2-F5 (T30 toroids)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 6/6


USB-C input + ESD
J12
USB_C_Receptacle
+5V PWR_FLAG
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
R30
5.1k
R31
5.1k
USB_DM
USB_DP
SH
A1
PWR_FLAG
D9
SP0503BAHT
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
+5V
PWR_FLAG
C31
4.7u
C30
100n
C32
100n
C33
4.7u
R32
1k
R33
22.1k
R34
47.5k
RST
VREGIN
VDD
VIO
VBUS
D-
D+
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
U10
CP2102N-A02-GQFN24
+3.3 V regulator (TLV75533, 500 mA)
U11
C34
1u
C36
4.7u
IN
EN
OUT
NC
TLV75533PDBV
TLV755P DS (Table 4-1, ROC): CIN, COUT >= 1 uF ceramic (>= 0.47 uF effective),
COUT <= 200 uF. EN tied to IN. Dropout <= 238 mV @ 500 mA.
IQ 25 uA, PSRR 46 dB @ 100 kHz. Pin 4 NC (LP2985 BP cap C35 removed).
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5
(EtherKeyerMini pinout):
1 VCC ref only, 2 TXD,
3 RX/UPDI, 4 GND.
No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
UPDI_RX
R35
470
R37
10k
C37
100n
R36
1k
DTR
RESET
MCU (AVR128DA48, DxCore, 3.3 V)
C39
1u
C49
100n
C38
100n
C40
100n
U9
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
PF2
PF3
PF4
PF5
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
PE0
PE1
PE2
PE3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
AVR128DA48-E/PT
C41
15p
C42 32.768kHz
15p
GPS_PPS
BTN_DSP_2
D38_A20_PF4
D39_A21_PF5
D22_A0_PD0
D23_A1_PD1
D24_A2_PD2
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
BTN_DSP_1
D31_A9_PE1
D32_A10_PE2
D33_A11_PE3
MCU_TX
MCU_RX
SDA
SCL
TX
BAND_A
BAND_B
D7_PA7
GPS_RXD
GPS_TXD
CAL_CLK
SI_INTR
OLED_DC
OLED_RES
UART_TX
UART_RX
SYNC_LED
D17_PC3
OLED_MOSI
D19_PC5
OLED_SCK
OLED_CS
SYNC_LED
R38
470
D11
SYNC
TX
R39
470
D12
TX
VERIFY
[ ] U9 AVR128DA48 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10), ADC PD0-PD2,
BAND_A PA5 / BAND_B PA6, D4 PA7, SYNC LED PC2, GPS USART3 DEFAULT PB0 TX / PB1 RX; firmware must follow.
[x] GPS_PPS on PF2 (fully async pin); EVSYS CHANNEL4/5 gen 0x4A -> USERTCBOCAPT (DS 21.5, 16.3.4).
[ ] CAL_CLK (SI5351C CLK4 via R47 33R) on PB2 (async, EVSYS CH0/1 gen 0x4A) -> TCB0 COUNT;
TCB0 OVF -> TCB1 COUNT (CASCADE), PPS CAPT on both = 32-bit count per 10 s gate (DS 22.3.3.3).
[ ] SI_INTR (SI5351C INTR, open drain, R46 10k pull-up) on PB3; BTN_DSP_1 on PE0.
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0);
trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4); ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6/Q7 AO3400A drive the 6-relay tree (Q6 2 coils ~42 mA, Q7 4 coils ~84 mA); Q8/Q9 removed.
[ ] +5V now from USB-C VBUS only (500 mA default USB); PA draw not yet measured.
[x] U1 FB tied to VIN: Jason: FB tied to VIN selects internal 12 V output (TLV61046A datasheet).
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AVR128DA48 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 - MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 4/6



VERIFY

- [x] Antenna short: TPS22941 limit 40–80 mA, auto-restart, avg <= 27 mA (was 320 mA).
- [x] V_BCKP: CR1220 in KH-BS1220-2-SMT (C5365932): C47/D13/R41 removed.
- [ ] BT1 footprint: draw from KH-BS1220-2-SMT DS.
- [ ] U13 TPS22941 JLC stock thin (26 pcs).
- [ ] Short peak 80 mA > VCC_RF 50 mA op. (DS T13) for <= 20 ms bursts; confirm on bench.
- [ ] V_IO ramp 25–35000 us/V (DS 4.1): measure.
- [ ] GPS_PPS MCU pin: see mcu_usb.

RECEIVER: VCC + V_IO tied to +3.3V; VIO_SEL open = 3.3 V I/O (IM 4.1.2).
C43/C44 beside pins 7/8, short GND vias; no series R > 0.2 ohm in VCC (IM 4.1.1).
RESET_N, EXTINT, SCL, SAFEBOOT_N, LNA_EN left open (DS Table 10).
SDA = ANT_SHORT_N (IM 3.3.1.2): CFG-HW-ANT_CFG_SHORTDET=1, ANT_SUP_SHORT_PIN=SDA PIO.
PPS: TIMEPULSE = SAFEBOOT_N via 1k inside (DS fn 15); nothing may hold it low at power-up, so no pull-down. R42 limits a shorted J14 to ~7 mA (I_PIO 10 mA).

ANTENNA: RF_IN is DC-blocked inside the module (IM Table 1), no external DC block.
Bias-T VCC_RF -> U13 TPS22941 -> R40 10R -> C45 10n -> L3 27 nH onto RF line (IM 3.3.1.2, 4.3.4).
U13: I_LIM 40–80 mA, 5–20 ms blanking, 40–160 ms restart (TPS2294x DS 7.5): short avg <= 27 mA, typ 7 mA (VCC_RF abs max 250 mA, DS Table 12). 25 mA antenna drop ~0.27 V -> ~2.9 V at J15.
Layout: 50 ohm CPWG pin 11 -> J15, L3 pad on the line, >= 5 mm from digital.
BACKUP: BT1 CR1220 (3 V, 35 mAh, user-fitted, NOT rechargeable, no charger) on V_BCKP;
C46 1u at pin 6 (IM 4.1.3). 35 mAh / 28 uA (DS Table 17) = ~1250 h (~52 days) powered off.

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MAX-M10S on USART3 (PB0/PB1), PPS to J14, active-antenna bias-T, CR1220

Etherkit

Sheet: /GPS/
File: gps.kicad_sch

Title: OpenBeacon 2 – GPS (u-blox MAX-M10S)

Size: A4	Date: 2026-10-25	**Rev: C**
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